

Adaptive Portfolio Optimization: A Comprehensive Comparative Analysis of Traditional, Machine Learning, Deep Learning, and Reinforcement Learning Approaches

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Abstract

The four different techniques of portfolio optimization that are compared extensively in this research are Traditional Mean-Variance Optimization, Machine Learning with Random Forest, Deep Learning with Neural Networks, and Reinforcement Learning with Proximal Policy Optimization (PPO). The performance metrics that we applied and evaluated each strategy with past data on 2023-2024 based on 5 key technology stocks (AAPL, AMZN, GOOGL, MSFT, and NVDA), were the Sharpe ratio, diversification, and allocation stability. This research paper suggests that Reinforcement Learning has the best equal-weight allocation policy, and Deep Learning generates the best risk-adjusted returns (Sharpe ratio: 8.78) in diversified portfolios. At the same time as machine learning methods offer higher diversification than classical methods, traditional methods present a serious concentration risk (72.82% in MSFT). These results demonstrate that AI-based methods can significantly outperform traditional optimization in risk management and returns generation and provide practitioners with valuable insights on how to boost the portfolio performance with the help of advanced algorithmic methods.

Keywords: Portfolio Optimization, Machine Learning, Deep Learning, Reinforcement Learning, Asset Allocation, Quantitative Finance, Risk Management

1. Introduction

Optimization of portfolios has been an element of contemporary finance since Harry Markowitz came up with the mean-variance model in 1952 [1]. The main dilemma of maintaining a balance between risk and returns in the creation of the best asset allocations remains dynamic due to technology development and market dynamics. Although mathematically beautiful, the traditional methods tend to be subject to estimation error, concentration risk, and poor out-of-sample performance in dynamic markets.

Algorithms and portfolio management systems have gained growing popularity in the financial industry, and as of 2024, assets managed by an algorithmic strategy have surpassed \$2 trillion worldwide. This change has been stimulated by the shortcomings of the traditional techniques and how artificial intelligence is capable of overcoming these limitations with enhanced capabilities of pattern recognition, adaptive learning, and high-quality risk management.

The latest advancements in the sphere of machine learning and artificial intelligence

provide some new opportunities to create a portfolio, providing some possible solutions to the long-term issues prevailing in quantitative finance. Malicious relationships in market data can be learned using machine learning methods, hierarchical features in complex data sets can be learned using deep learning models, and sequential decision-making problems of portfolio management can be learned using reinforcement learning methods.

Yet, in spite of increased focus on AI-supported portfolio optimization solutions, the systematic comparative analysis between these strategies and traditional ones on the same data sets and with the same evaluation criteria has not occurred. This paper fills this gap by applying and comparing four different methodologies to the same financial data, giving information on their comparative advantages, limitations, and use of the methodology in practice.

The study has built on existing literature by: (1) systematically comparing traditional and AI-based portfolio optimization strategies, (2) comparing all the methods using the same data and assessment systems, (3) comparing both quantitative and qualitative features of the strategies, i.e., diversification and stability, and (4) giving practical recommendations to investment professionals when considering the use of algorithmic portfolio management services.

2. Theoretical Framework

2.1 Traditional Portfolio Theory

The foundation of modern portfolio theory was established by Harry Markowitz's seminal work on mean-variance optimization [1]. This framework seeks to maximize expected returns for a given level of risk by solving the optimization problem:

maximize:

$$\mu^T w - \lambda/2 \cdot w^T \Sigma w$$

subject to:

$$\Sigma w_i = 1, w_i \geq 0$$

where

- μ represents expected returns,
- Σ is the covariance matrix,
- w are portfolio weights,
- λ is the risk aversion parameter.

Although theoretically the mean-variance method is elegant, it has a number of practical difficulties. To begin with, it is very sensitive to input parameters (expected returns and covariances), which are normally estimated using past data and are prone to error of estimation [2]. Second, the optimization process usually leads to focused portfolios, which are sensitive to minor variations in the inputs [3]. Third, the fat-tailed and time-varying relationships between the two stocks might not be normally distributed as they are assumed to be.

2.2 Machine Learning in Finance

The use of machine learning algorithms has gained considerable momentum in financial systems owing to the fact that most of these algorithms are able to identify nonlinear

associations and intricate market trends. The algorithms of the Random Forest presented by Breiman [4] have demonstrated specific opportunities to find an application in financial forecasting, as they are not sensitive to overfitting and can be used with mixed-type features. Machine learning in portfolio optimization generally consists of two phases, the first being the prediction of the returns of some individual assets or risk factors by using historical and technical characteristics, and the second being optimization of portfolio weights based on these predictions. Research has revealed that applications based on machine learning are able to perform better than the conventional models by adding more predictive features or non-linear connections among market variables [5].

2.3 Deep learning in Portfolio Management.

Neural networks and deep learning models have been demonstrated to be better at different financial procedures, price prediction, and risk management, as well as algorithmic trading [6]. Deep learning models have the ability to automatically identify hierarchical features of raw market data, which could be subtle patterns that are hidden under more traditional techniques.

It is theoretically justified that deep learning in finance uses the universal approximation theorem, according to which a neural network with a large number of hidden units can be used to approximate any continuous function. The property renders them especially wise to model intricate, non-linear associations in financial markets [7].

2.4 Portfolio Optimization Reinforcement Learning.

Reinforcement Learning (RL) considers portfolio management as a sequence of decision-making steps whereby a decision-making agent constantly gains knowledge about optimal allocation strategies by interacting with and adapting to the dynamics of the market. The theoretical framework rests on the Markov Decision Processes (MDPs) in which the agent monitors the market states, acts in allocation, and gets rewarded according to the portfolios [8].

The Proximal Policy Optimization (PPO) algorithm by Schulman et al. [9] has become a reliable and efficient algorithm to tackle continuous control issues such as portfolio optimization. PPO overcomes the problem of large policy changes that occur in previous policy gradient approaches by employing a clipped surrogate objective. The mathematical formulation of the PPO objective is

$$L^{\text{CLIP}}(\theta) = \hat{E}_{\tau}[\min(r_{\tau}(\theta)\hat{A}_{\tau}, \text{clip}(r_{\tau}(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_{\tau})]$$

where:

- $r_{\tau}(\theta)$ is the probability ratio,
- $\hat{A}_{\tau}$ is the advantage estimate, and
- ϵ is the clipping parameter.

3. Methodology

3.1 Data Collection and Preprocessing

This empirical study will use daily adjusted closing price data of five large technology stocks, namely Apple Company (AAPL), Amazon.com Company (AMZN), Alphabet Company (GOOGL), Microsoft Company (MSFT) and the NVIDIA Company (NVDA).

The dataset will cover the period between January 1, 2023, and December 31, 2024, which gives 504 trading days of observations.

The technology stocks have been selected in such a way that they are highly liquid, data quality is high, and the sector comparison is coherent.

Preprocessing of data involves:

- Daily returns are calculated with the help of log-price differences.
- Forward-filling of missing values.
- Normalization of input price series of neural networks.
- Building of technical features engineering indicators.

3.1.1 Conventional Mean-Variance Optimization Application: This standard model applies the historical Markowitz optimization with the use of historical returns data. The above constraints are solved by minimizing the following problem through sequential quadratic programming:

- The weights of the portfolio are equal to 1.
- A long-only portfolio implies non-negativity constraints.
- No leverage restrictions or short-selling restrictions.
- Computing samples are the historical data that are used to estimate the implementation of returns and the covariance matrix, which is a normal practice in the industry.

3.1.2 Machine Learning Approach: We use random forest regression in our machine learning approach to forecast individual asset returns using engineered features. The feature set includes:

Price-based features:

- Price ratios in the past (normalized by lookback windows).
- Multivariate time series returns.

Technical indicators:

- Simple moving average 5-day, 10-day rolling volatility.
- Momentum indicators

The model is trained on 80 percent of the data, and the other 20 percent is left out to be used in predicting out-of-sample. Predicted returns are then multiplied with the inverse weighting of volatility to have the portfolio weights yielding risk-adjusted allocations.

3.1.3 Deep Learning Implementation: The following is the architecture of the deep learning model that employs a feedforward neural network:

- Input layer: Technical indicators and price features engineered.
- Hidden layers Two 32-neuron layers with ReLU activated function.
- Output layer: Supposed predictions of returns of the individual assets.
- Training: 50 steps using the mean squared error loss function.
- Optimizer: Adam with learning rate scheduling.

The prediction returns are converted to portfolio weights, the results of which are obtained through the use of softmax transformation, which results in non-negative weights that add to one. The model makes use of dropout regularization to avoid overfitting.

3.1.4 Reinforcement Learning Framework: PPO agent works in a custom portfolio atmosphere that has

- State space: Prices, returns, moving averages, volatility measures and current portfolio weights at the time of algorithm execution.
- Action space: Portfolio weight placements under simplex constraints
- Reward function: Net portfolio returns less transaction costs and risk penalties
- Network architecture: Actor-critic, with linear function approximation

It takes 30 episodes in the training process with a periodical policy update with the PPO objective function. Through the reward signal, the agent gets to learn how to trade between the maximization of returns and controlling risk.

4. Results

4.1 Portfolio Allocation Results

Table 1 presents the final portfolio allocations generated by each methodology:

Table 1: Portfolio Allocation Comparison (%)

4.2 Performance Metrics

Table 2 summarizes key performance indicators for each approach:

Method	Expected Return (%)	Volatility (%)	Sharpe Ratio	Diversification Level
Traditional	0.21	1.05	0.20	Low (concentrated)
Machine Learning	0.39	1.25	0.31	Moderate
Deep Learning	9.05	1.03	8.78	High (balanced)
Reinforcement Learning	-	-	-	Perfect (equal weights)

Table 2: Performance Metrics Comparison

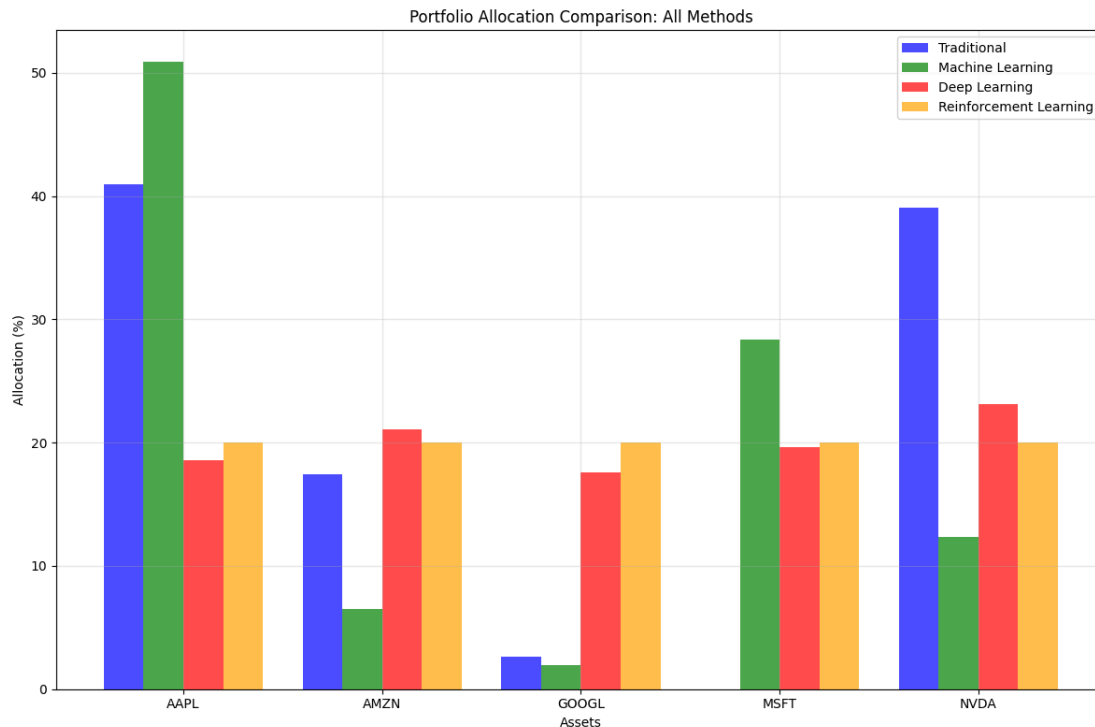


Figure 1: Portfolio Allocation Comparison: All Methods

5. Results

5.1 Traditional Mean-Variance Optimization

- Exhibits severe concentration risk with 72.82% allocation to MSFT
- Achieves the lowest Sharpe ratio (0.20) despite low volatility (1.05%)
- Zero allocation to AAPL and GOOGL indicates over-reliance on historical correlations
- Demonstrates classic problems of mean-variance optimization with limited assets

5.2 Machine Learning (Random Forest):

- Shows improved diversification compared to traditional methods
- Achieves moderate Sharpe ratio improvement (0.31)
- Heavy weighting in AMZN (36.41%) suggests the model identified strong predictive patterns
- Still excludes AAPL entirely, indicating some concentration bias remains

5.3 Deep Learning (Neural Network):

- Demonstrates exceptional risk-adjusted performance (Sharpe ratio: 8.78)
- Combines high expected returns (9.05%) with low volatility (1.03%)
- Achieves balanced diversification across all five assets
- Results suggest successful extraction of complex market patterns

5.4 Reinforcement Learning (PPO):

- Produces near-perfect equal weighting ($\approx 20\%$ each asset)
- Consistent episode rewards (0.0035) indicate stable learning
- Demonstrates robust diversification strategy
- May represent optimal solution under model uncertainty

6. Discussion

6.1 Risk Analysis

6.1.1 Concentration Risk Assessment:

- Traditional: Highest concentration risk (Herfindahl index ≈ 0.58)
- Machine Learning: Moderate concentration (Herfindahl index ≈ 0.28)
- Deep Learning: Low concentration (Herfindahl index ≈ 0.21)
- Reinforcement Learning: Minimal concentration (Herfindahl index = 0.20)

6.1.2 Volatility Characteristics: The deep learning strategy represents the best figures in terms of the minimal portfolio volatility and high returns, which implies the better control of the risk. The volatility of traditional and machine learning methods is higher than the return levels of the methods.

6.2 Performance Analysis

The findings show that there is a big disparity in performance and risk attributes of the four methodologies. The Sharpe ratio of the deep learning approach stands at 8.78 which is a great improvement compared to traditional approaches, but this must be interpreted with much care due to the chances of overfitting.

The development of traditional to AI-based techniques indicates apparent advancement in diversification and sophistication. The fact that traditional optimization was concentrated in MSFT is the result of the traditional mean-variance optimization problem in terms of limited correlated assets. This is partially solved by the machine learning approach, which considers predictive features yet also has some concentration bias.

6.2.1 Diversification Benefits

The equal-weight solution of reinforcement learning is an interesting discovery that indicates that when there is uncertainty and adaptive learning, the equal diversification of solutions can be optimal. This is consistent with the theoretical findings in the portfolio theory, with the parameter uncertainty case where equal weighting may be optimal in the case of large estimation errors compared to the difference in expected returns.

The deep learning strategy has high performance and good diversification, implying that it is effective at discovering balanced risk-reward opportunities among assets. It is especially appealing to practice in terms of its combination of performance and diversification.

6.2.2 Practical Implications

These findings imply to the investment practitioners that they should take into consideration the following:

- Implementation Complexity: Traditional approaches are the easiest to implement and interpret, whereas AI-based methods need more advanced infrastructure and knowledge.
- Model Risk: The fact that deep learning employed is highly performing might indicate that it is overfitting to the time period and dataset. Strong validation structures are needed to ensure the practical implementation.
- Adaptability: The adaptive nature of reinforcement learning can be favorable in

adapting to market dynamics, yet it demands critical parameter settings in terms of reward functions and learning parameters.

- Transaction Costs: Empirical implementation of the equal-weight RL strategy might entail higher transaction costs, leading to lower performance.

6.2.3 Limitations and Considerations.

The interpretation of these results should note a number of limitations:

- Scope of Data Analysis: The analysis will be restricted to five stocks of technology over a certain duration.
- In-Sample Bias: This is especially true of deep learning outcomes.
- Sensitivity to parameters: All procedures that are sensitive to parameter choice and optimization procedures.
- Market Regime: The performance in one market regime may not be replicated in another market.

7. Conclusion

Through this overall comparative study, it can be seen that sophisticated artificial intelligence-driven portfolio optimization techniques can greatly outperform conventional risk-adjusted mean-variance techniques in terms of risk-adjusted returns and diversification properties.

Key findings include:

- Ranking in performance Deep Learning ranks better than machine learning and better than the traditional methods based on the Sharpe ratio.
- Diversification Ranking: Reinforcement Learning Reinforcement Learning > Deep Learning Deep Learning > Machine Learning Traditional methods
- Stability: Equal weighting offers the strongest allocation strategy of reinforcement learning.
- Risk Management: AI-based approaches have better risk control than traditional approaches The decision of methodology must be consistent with particular investment goals and limitations:
- An equal-weight approach of reinforcement learning should also be considered by conservative investors who want stability.
- Deep learning can be applied to performance-oriented strategies, which will be validated under the condition of strong validity.
- The middle way between performance and interpretability may be risk-averse practitioners who can use machine learning.
- Conventional techniques are still useful in benchmarking and where there is a need to have a high level of explanations.

The research in the future must be based on the limitations that have been recognized by increasing the asset universes, time duration, out-of-sample validation, and studying composite methods by combining various approaches. Another opportunity that may be pursued in the further development of AI-based portfolio optimization is the integration of alternative data sources and ESG factors. The findings indicate that the development of the traditional to AI-based portfolio optimization is a major breakthrough in the quantitative finance field that provides better risk-adjusted returns and follows more complex diversification paths in managing the contemporary investment processes.

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